Lab 1

Introduction to Programming - Lab 1

Vietnamese-German University
Instructor: Dr Tran Huu-Tam
Teaching assistants: Trung-Son and Anh-Hieu

Student ID: 10426086

Answer the theory questions. Follow the instructions in the Practical section for the programming exercises.

Theory

Question 1: Reproducible execution in an editor

A student edits one file but repeatedly runs an older file with the same name in another folder. Explain how to check the saved file, its path and the selected Python interpreter. Describe a simple change that would confirm which file actually ran.

Question 2: Printing a rectangular pattern

Describe how to print three rows of four `*` characters using nested loops and `print(..., end="")`. Explain where the newline-producing `print()` belongs. What display mistake occurs if it is indented inside the inner loop instead?

Practical

Submission: provide one complete Python 3 script for each assigned exercise, together with your test cases. Explain important decisions as code comments. Screenshots are not required.

Scope: use variables, scalar values, strings, arithmetic, conditions, and `for` or `while` loops. Built-in input/output, conversion, `range`, `len`, `abs`, `min`, `max`, and basic string operations are permitted. Do not define functions or classes, import libraries, or use lists, tuples, dictionaries or sets. Algorithmic reasoning is allowed in this Practical section.

Input and output: each numeric value is provided on its own line, in the order stated in the question. A word, command or string occupies one whole line. Print only the requested result, without input prompts. In compact examples, `/` marks a line break; the slash is not part of the actual input or output. Test at least one boundary case and one ordinary case in addition to the example. Multiple valid outputs are accepted only where explicitly stated.

Device: Computer (Windows)
Browser ID: 2df4aba9-6fc6-4882-a67f-5390b0e52bc1
Connection IP: 113.161.251.188
Downloaded at 2026-10-14 14:22:48 UTC+07:00 (Asia/Saigon)

Question 3: The single parity exception

Exactly one input integer has parity different from all the others. Find its position, counting from one. Process values as they arrive and retain counters and relevant positions rather than a list of all numbers.

Input: Read n, then n integers.

Output: Print the exceptional position.

Bounds: 3 <= n <= 1000; values are 1 through 1000.

Example: input `5 / 2 / 4 / 7 / 8 / 10`; output `3`.

Question 4: As many prime pieces as possible

Represent n as a sum of the largest possible number of prime integers. A prime is greater than one with exactly two positive divisors. Print a valid representation. You may reason about small primes rather than constructing a table of primes.

Input: Read n.

Output: Print the number of addends, then one addend per line. Any optimal representation is accepted.

Bounds: 2 <= n <= 1000.

Example: input `7`; output `3 / 2 / 2 / 3`.

Device: Computer (Windows)
Browser ID: 2df4aba9-6fc6-4882-a67f-5390b0e52bc1
Connection IP: 113.161.251.188
Downloaded at 2026-10-14 14:22:48 UTC+07:00 (Asia/Saigon)